

COMPUTER ARCHITECTURE

EARL KINUTHIA – SCT212-0067/2019

Summary Notes

CHAPTER 1

FUNDAMENTALS OF QUANTITATIVE DESIGN AND ANALYSIS

1.1 INTRODUCTION

Overview of Technological Progress

Computer technology has made great strides in the last 65 years, resulting in the affordability and accessibility of high-performance computers.

Both new computer architectures and technical developments have improved performance.

The Revolution of Microprocessors

Performance increase was further accelerated with the introduction of microprocessors in the late 1970s, reaching about 35% annually.

The advent of standardized operating systems like UNIX and Linux, as well as the fall of assembly language programming, are major causes behind this change.

Impact of RISC Architecture

Computer design was completely transformed in the early 1980s with the advent of this architectures.

The architecture's simplified design and improved performance by optimizing cache usage and instruction-level parallelism.

RISC-like approaches were internalized by Intel, allowing x86 architectures to compete with RISC designs.

Performance Growth and Influence

Advances in technology together with architectural innovation led to an unprecedented 50% growth in performance annually during the 1980s and early 2000s.

This growth allowed for the creation of new classes of computing systems, such as warehouse-scale computers, smart devices, and personal computers.

Evolution of Software

Evolutionary developments in performance enabled programmers to concentrate on productivity instead of sheer performance.

This led to the rise in popularity of Java and C# programming languages as well as Ruby and Python scripting languages. Cloud computing and Software as a Service models equally transformed software deployment.

Challenges and Future Directions

Advances in single processor performance after the year 2003 reduced to around 22% annually due to power consumption and the demise of instruction level parallelism.

The created a shift toward data level, thread level and request level parallelism to achieve higher gains in performance.

Future architectures require explicit parallel programming, introducing new challenges for software developers.

1.2 CLASSES OF COMPUTERS

Transformation in Computing

The evolution of computing has revolutionized how we utilize technology, and this has created phenomenal growth in computer applications and markets. Computing has become more specialized over time, and five general categories of computing have been established.

1. Personal Mobile Devices

These are smartphones, tablets, and other wireless devices that are multimedia-capable and are designed with the following significant factors in mind:

Cost-effectiveness These consumer devices need to be low-cost, ranging from a few hundred to a thousand dollars.

Energy efficiency Given that they rely on battery power, energy management is crucial. Efficient power usage extends battery life and reduces heat dissipation.

Storage preferences Instead of traditional hard drives, they use flash memory which is compact, durable, and power-efficient.

Real time performance Most of their applications such as video playback and online gaming, require strict real time processing for responsiveness.

Memory optimization Since they have limited resources, software and applications need to be built to maintain low memory usage while maintaining performance.

2. Desktop Computing

Desktop computing has been a top computing category, offering a balance of price and performance. Main features are:

Variety in price range Desktop computers range from budget laptops to professional workstations, typically between \$300 and \$2,500.

Optimization of performance These computers aim at computing and graphics performance, hence suitable for a variety of applications, varying from regular usage to intensive gaming and professional usage.

Technological innovation Desktop computing is a proving ground for new microprocessor architectures and high-end hardware components.

Performance issues with web-focused applications With more use of cloud-based applications and web services, desktop performance measurement is shifting to encompass network efficiency and browser optimization.

3. Servers

Servers are a part of large scale computing environments that provide robust and stable services to organizations and businesses. Their defining features are:

High availability Servers operate continuously to handle services such as online banking, e-commerce and cloud computing.

Scalability As an increasing number of services are demanded, server design must scale effectively in memory, processing power, and storage capacity.

Throughput and efficiency Compared to desktops for personal performance, servers focus on handling large numbers of transactions per second with optimal efficiency.

4. Cluster Computers

The development of cloud computing and big data processing has led to warehouse-scale computers, which are gigantic clusters of connected systems. Their key characteristics are:

Cost and power efficiency As these facilities require massive resources, minimizing power consumption and cooling costs is a top priority.

Redundancy and fault tolerance They rely on inexpensive components designed to fail gracefully, while detection and isolation of faults using software enable fault-free operation.

Comparison to supercomputers Unlike traditional supercomputers grounded on high speed floating point computation, they focus on the handling of huge amounts of internet traffic, storage, and interactive jobs.

5. Embedded Computers

Embedded computers are special purpose computing hardware embedded within appliances and industrial devices. They contrast with general-purpose computing hardware in several aspects:

Diverse performance levels Embedded systems range from low-performance 8-bit processors for household appliances to high-performance processors in networking devices.

Cost-oriented design Contrary to other segments of computing, in which performance is a focus area, embedded computers focus on providing required functionality at minimum cost.

Limited software flexibility Unlike personal mobile devices, embedded systems usually execute fixed software and do not readily support third party applications.

Parallelism in Computing

Parallelism is now a part of modern-day computing due to the growing demand for computational power. The two main types of parallelism include:

Data-Level Parallelism This is the execution of one operation on various data elements simultaneously. It is often used in vector processing and multimedia.

Task-Level Parallelism This is the execution of many independent tasks in parallel to one another e.g multiple programs being executed simultaneously on a multi-core processor.

To achieve these forms of parallelism, four main hardware techniques have developed:

Instruction-Level Parallelism Pipelining and speculative execution allow one processor to execute multiple instructions at the same time.

Vector Architectures & GPUs These architectures exploit single Instruction, multiple data processing to execute the same instructions on many points of data in parallel.

Thread-Level Parallelism Multi-core and multi-threaded processors enable multiple threads to run at the same time, improving performance for multi-tasking applications.

Request-Level Parallelism Large-scale distributed computing systems such as clusters run independent requests in parallel, enhancing efficiency in cloud computing.

Flynn's Taxonomy of Parallel Architectures

Computer architectures can be classified into four classes based on their instruction and data handling capabilities:

Single Instruction, Single Data This is a traditional uniprocessor approach where a single instruction is executed on a single data element at a time.

Single Instruction, Multiple Data A single instruction operates on multiple data in parallel, such as in vector processors and GPUs.

Multiple Instruction, Single Data This a rare architecture where multiple instructions operate on the same data stream.

Multiple Instruction, Multiple Data This is the most general and most popular parallel architecture, which supports both task-level and data-level parallelism.

1.3 DEFINING COMPUTER ARCHITECTURE

The Role of Computer Architects

Computer architects are faced with designing computers that are performance-focused, power-sipping, cost-conscious and power-conscious. The process is multi-faceted and involves instruction set design, functional organization, logic design and hardware implementation. Implementation is succeeded by circuit design, power management and packaging. An architect must be good at software (compilers and operating systems) and hardware (logic design and physical packaging).

Evolution of Computer Architecture

Historically, instruction set design had been essentially computer architecture, but modern computer design is much more than that. While Instruction Set Architecture remains a critical component, the overall subject now includes processor organization, performance, and system integration.

Instruction Set Architecture:

ISA defines how software interacts with hardware e.g. the instruction set a processor supports. The most important characteristics of an ISA are:

1. Class of ISA

The majority of modern ISAs use a general-purpose register architecture where operands are in registers or memory. Examples include:

80x86 16 general-purpose registers.

MIPS 32 general-purpose and 32 floating-point registers.

ARM A load-store architecture, similar to MIPS, with explicit load and store instructions for accessing memory.

2. Memory Addressing

Modern ISAs use byte-addressable memory. ARM and MIPS need memory alignment for efficiency, whereas 80x86 has unaligned access with efficiency compromises.

3. Addressing Modes

Instruction designation of memory positions varies between ISAs. MIPS uses register, immediate and displacement addressing. ARM has PC-relative and autoincrement/autodecrement addressing modes. 80x86 offers more involved modes like base-plus-index and scaled addressing.

4. Operand Sizes and Types

Most ISAs support a range of operand sizes like 8-bit (ASCII characters), 16-bit (Unicode), 32-bit (integers), 64-bit (double words) and IEEE 754 floating-point formats.

Instructions Instruction sets have typically had data transfer, arithmetic/logical, control flow, and floating-point instructions. MIPS is a good representative of a small and efficient RISC ISA, while 80x86 has a broader and more complex instruction set.

6. Control Flow Instructions

They are returns, procedure calls, and conditional branches. MIPS places return addresses in registers, but 80x86 keeps them in memory. Branch execution relies on condition codes or register comparisons being used.

7. Instruction Encoding

ISAs can use fixed-length (MIPS, ARM) or variable-length (80x86) instruction formats. Fixed-length formats ease decoding, while variable-length formats conserve code space at the cost of making it more difficult to execute.

1.4 TRENDS IN TECHNOLOGY

The Imperative for Technological Adaptation in Computer Design

For a computer instruction set architecture to be up-to-date, it must adapt to fast-paced technological evolution. A sound ISA design could endure for years, like the mainframe ISA of IBM which has been consistently used for nearly 50 years. To qualify for long use, computer designers must plan and incorporate emerging technological trends into the design.

Core Implementation Technologies That Define Computer Architecture

There are five core areas of technology that impact modern computer implementations, with each having an extremely high growth rate:

1. Integrated Circuit Logic Technology

Transistor density grows at a rate of about 35% per year, leading to a 40%-55% increase in transistors yearly (doubling roughly every 18 to 24 months).

Referred to as Moore's Law, this trend enables continued growth in processing power and efficiency.

2. Semiconductor DRAM (Dynamic Random-Access Memory)

DRAM capacity per chip rose by 25%-40% per annum, approximately every two or three years.

Trends have fallen behind and there is only advance possible if one switches to new technology because miniaturization potential is restricted.

3. Flash Semiconductor Memory

Widely used in cellular phones, Flash memory has been growing at 50%-60% per year, doubling every two years.

Flash remains much cheaper than DRAM (by a factor of 15-20 times per bit) but slower.

4. Magnetic Disk Technology

Density growth rates varied historically but are currently at 40% per year, doubling every three years.

Disks remain the most affordable mass storage, 300-500 times cheaper per bit than DRAM.

5. Network Technology

Improvements are decided by both switch performance and transmission speed, enabling rapid data transfer between computing infrastructures.

Modern systems require high-bandwidth interconnects in order to keep pace with computation and communication rate.

Performance Trends: Bandwidth vs. Latency

Bandwidth is the amount of work completed in unit time e.g. MB/s of disk transfer while latency is time elapsed from initiation of request to completion e.g. milliseconds of disk access.

Microprocessors and networks have seen greatest improvements:

- 10,000-25,000× bandwidth increases.
- 30-80× latency reductions.

Disk and memory technologies have maximized capacity at the cost of performance, while bandwidth still increased 300-1200×, with latency gains being 6-8×.

Bandwidth advancement exceeds latency advancement in all technologies, and a rough rule of thumb is that bandwidth improves at least as the square of latency improvements.

Transistor Scaling and Wire Performance Challenges

Feature size of integrated circuits has come down substantially, from 10 microns in 1971 to 32 nanometers in 2011, with 22-nanometer chips being on the production lines.

The density of transistors grows quadratically, enabling processing capacity to increase exponentially, making possible shifts from 4-bit to 64-bit processors and now supporting multicore designs.

The performance of wire does not keep pace as much as that of transistors, creating new challenges:

As wires become smaller, wire resistance and capacitance are greater, making signal propagation take longer.

Even though wires are shorter, wire transmission delays now occupy more of the processor's clock cycle.

Wire delay is now a dominant design constraint, in competition with power dissipation as a significant limiting factor.

It is in understanding these trends that computer architects are able to create systems which can handle abrupt technological developments. Optimizing ahead of time for future memory, processing, and interconnect technology, they enable longevity and productivity in computing design.

1.5 TRENDS IN POWER AND ENERGY IN INTEGRATED CIRCUITS

The Growing Importance of Power Management

Power consumption has become the main constraint in computer design for all computing categories. The issue is both power supply-sufficient supply without voltage drops and cooling-preventing overheating that could kill components. Sophisticated techniques are utilized by modern processors to manage power and provide stability in operations.

Key Power and Energy Considerations

Computer designers must address three major power aspects:

1. Maximum Power Requirements

Processors must fall within the range of the power supply system. Voltage sag due to excessive current draw can cause system failure.

2. Sustained Power Consumption

Thermal Design Power defines cooling requirements. Systems must be designed to handle average power dissipation as well as peak loads. To prevent overheating, processors dynamically reduce clock speeds as they approach thermal limits.

3. Energy Efficiency

In comparison to power (which calculates instantaneous usage), energy (measured in joules) measures the total usage over a period of time. Energy efficiency is a more reliable measure of processor comparison since it considers completion time of the workload as well as power consumption.

Dynamic Power and Voltage Scaling

In CMOS processors, dynamic power dissipation is predominantly determined by switching transistors. Power per transistor is directly proportional to the capacitance and the square of the operating voltage. This means:

Lowering voltage lowers power significantly.

Processor voltages have been lowered in the past 20 years from 5V to below 1V.

Greater numbers of transistors and higher frequencies of switching have resulted in raising total power consumption in spite of these voltage drops.

Methods of Power Efficiency

In addressing power constraints, today's processors employ a few strategies:

Clock Gating Idle processor units e.g. floating-point units & inactive cores see their clocks gate switched off for the purpose of saving power.

Dynamic Voltage-Frequency Scaling Processors alter voltage and frequency based on load level, acutely reducing power consumption during periods of idleness.

Design for Typical Case Memory and storage power-saving modes (e.g. DRAM low-power states, slower-spinning disks) extend battery life in mobile devices.

Turbo Mode Processors increase clock speeds temporarily when thermal conditions allow, increasing short-term performance without exceeding power limits.

Leakage Power Reduction As transistors are miniaturized, leakage current (power loss when a transistor is turned off) is a significant problem. New processors apply power gating to turn off unused parts of the chip entirely, minimizing static power losses.

Race-to-Halt Strategy More powerful processors, although less efficient, can process tasks in less time and enable the system to go into sleep mode earlier, conserving overall energy.

Since power is finite, measuring processor efficiency has shifted from performance per mm^2 of silicon to performance per watt (operations per joule). This has dramatically influenced parallel computing approaches, prioritizing efficiency over raw processing power.

1.6 TRENDS IN COST

The Growing Dominance of Cost in Computer Design

While cost is not crucial in some premium designs like supercomputers, cost-sensitive computing is gaining traction. The computing industry has leveraged technological innovations in the past 30 years to cut costs while getting more performance out of computers.

Computer designers must be knowledgeable about cost variables when deciding about the feasibility of new features, just like architecture builders must incorporate material costs.

Factors Contributing to Computer Costs

Computer prices are influenced by several key factors, including:

1. Time and the Learning Curve

Costs of production decrease with time due to gains in production efficiency and yield. It is directly reduced by yield gains in that higher yields result in fewer defective units. Prices of memory such as Dynamic random access memory price per megabyte, decrease steadily with time.

2. Volume and Mass Production

Higher production levels amplify learning curve benefits hence reducing per-unit costs. Mass production allows companies to defer development costs across high volumes of units which reduces per-unit costs. Rule of thumb applies a 10% cost reduction for every doubling of production volume.

3. Commoditization and Market Competition

Products that are becoming commodities such as flash memory see prices fall due to competition.

The personal computer market has evolved into a commodity market hence there is price reduction and diminishing margins.

Competition diminishes the gap between manufacturing price and selling price.

Cost of Integrated Circuits and Die Manufacturing

With increasingly more computing relying on integrated circuits, chip cost now accounts for a progressively larger system cost.

Important factors are:

1. Manufacturing Process & Die Yield

ICs are produced in wafers, which are tested and processed into functional chips (dies).

Yield improvements (number of functioning chips per wafer) substantially lower costs.

The size of the die controls the yield of good dies per wafer; smaller dies are higher yield and lower cost.

2. Die Size and Cost Sensitivity

The cost per chip increases roughly with the square of die size. Example: A 2.25 cm^2 die is actually $4 \times$ more expensive than a 1.00 cm^2 die, although it's only $2.25 \times$ larger. Designers can control die size by optimizing the number of I/O pins and reducing unnecessary features.

3. Redundancy to Improve Yield

In order to improve chip yield, modern designs incorporate redundant memory cells in Dynamic and Static RAMs and caches. Higher yield translates into lower overall production cost.

4. Fixed vs. Variable Costs

Large chips are aided by per-unit cost savings since fixed costs are spread across many units.

Low-volume chips have higher per-unit costs due to expensive mask set; new high-density chips involve \$1M+ masks, increasing prototype and production costs.

Some designs use reconfigurable logic or gate arrays in an effort to keep mask costs down for low volumes.

Cost vs. Price in the Computer Industry

- Margins between cost and selling price have contracted due to intense competition.
- Companies need to invest revenue in research and development, marketing, sales, manufacturing upkeep, and profit margins.
- Research and development spending varies by segment:
 - Commodity PC business account for 4% of revenue
 - High-end server business account for 12% of revenue.

Operational Costs in Large-Scale Computing

With cluster computing coming up, operational costs are just as significant as manufacturing costs:

- Server hardware and networking costs account for 60% of a data center's monthly cost (amortized 3–4 years).
- Power and cooling equipment account for 30% of operational cost (amortized 10 years).
- Energy efficiency has become a design priority, because cutting the power is an immediate cut in operating costs.